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Genetic diversity, population structure, and dispersal patterns of wild and cultivated *Cycas szechuanensis* in China

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ABSTRACT

Cycas szechuanensis (Cycadaceae) is an endemic and nationally protected species in China. To inform scientifically grounded conservation strategies of this rare and endangered plant, 19 wild and cultivated populations comprising 622 individuals from all known distribution localities were sampled, and their genetic diversity and population structure were assessed using microsatellite markers. Fine-scale spatial genetic structure (FSGS) and parentage analyses were conducted to evaluate pollen and seed dispersal in the two largest populations, Taishan and Yangjiang. Our results revealed high genetic diversity in *C. szechuanensis*, especially in wild populations. Bottlenecks were not detected in most wild populations, but inbreeding was evident. The persistence of cultivated populations primarily relied on clonal reproduction. Four genetic groups were identified in *C. szechuanensis*, corresponding to their geographic distribution from south to north, with the southern group exhibiting the highest genetic diversity. However, the greatest genetic differentiation was not observed between the southernmost and northernmost groups. Parentage analysis revealed restricted seed dispersal and long-distance pollen flow, with the farthest assigned paternity reaching 1.9 km, resulting in clustered recruitment and significant FSGS in juvenile individuals. Adult FSGS was not consistently significant after accounting for sex. Based on these findings, we make recommendations for *in situ* and *ex situ* species conservation.

1. Introduction

Cycads (Cycadaceae) are an ancient lineage of extant seed plants with a fossil record extending back to the Permian (Nagalingum et al., 2011). Despite their long evolutionary history, cycads are now among the most threatened plant groups globally. Approximately

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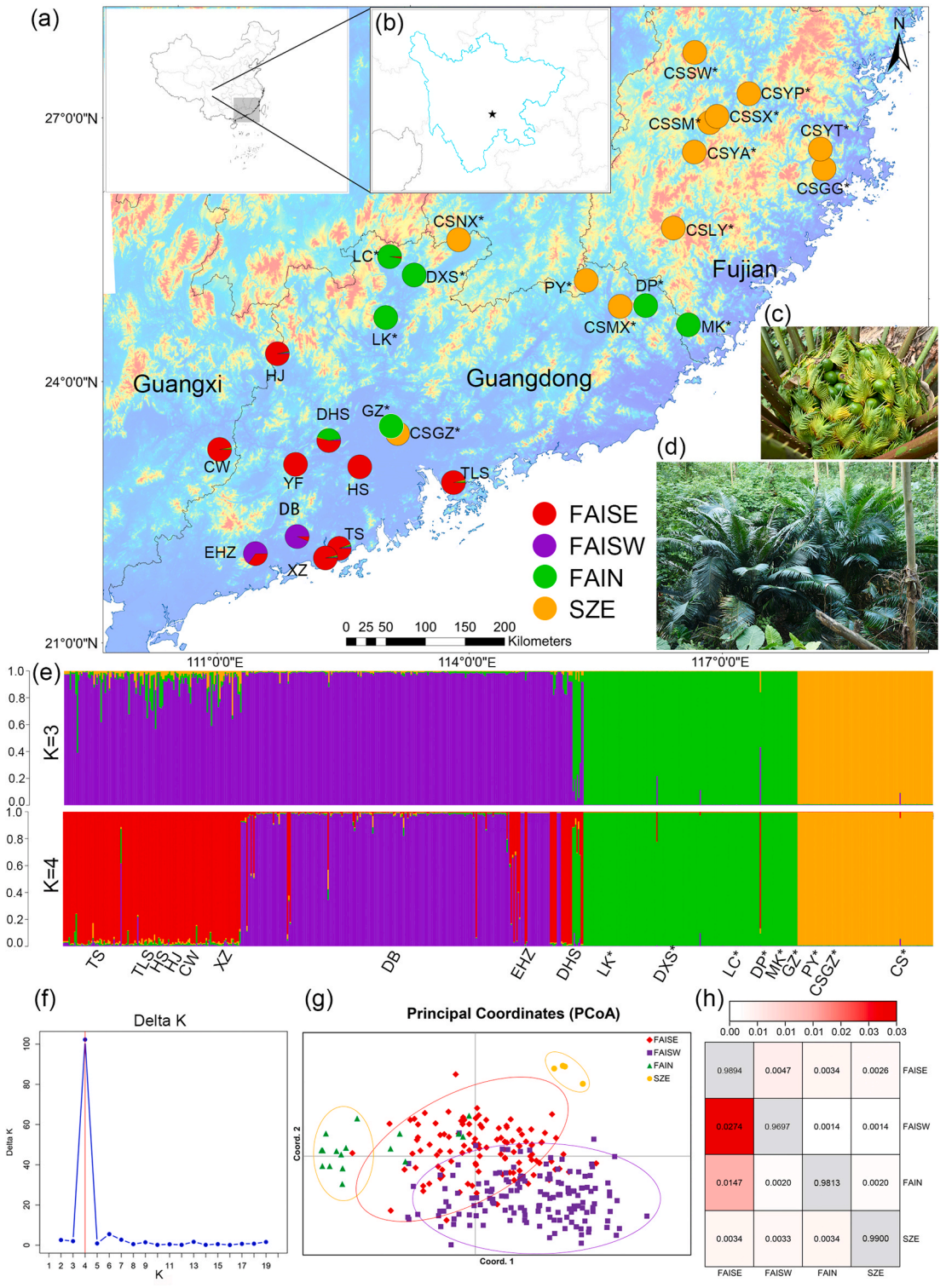


Fig. 1. Genetic structure of *Cycas szechuanensis* in China. (a) Sample collection localities and genetic structure of populations under $K = 4$ from STRUCTURE analysis. TS, DB, XZ, DHS, YF, and CW are newly recorded populations, and the remainder were documented in previous studies. (b) Holotype collection site of *C. szechuanensis*. (c) Female cone of *C. szechuanensis*. (d) Plants of *C. szechuanensis* in the wild. (e) Proportional membership of 19 *C. szechuanensis* populations based on $K = 3$ and $K = 4$ genetic clusters. (f) ΔK changes corresponding to K values from 1 to 20. (g) Genetic clusters across 19 *C. szechuanensis* populations based on PCoA. (h) Heatmap of recent migration rates calculated using BayesAss among four *C. szechuanensis* genetic clusters.

two-thirds of extant cycad species are threatened with extinction due to habitat destruction, overcollection for horticultural trade, and pollinator decline, leading to their recognition as a flagship group for plant conservation (IUCN, 2024).

The genus *Cycas*, comprising approximately 20 species, most of which are endemic, is the only cycad genus naturally distributed in China (Xi et al., 2022). Chinese *Cycas* species are predominantly distributed in the southwestern region, particularly Yunnan, Guangxi, and Sichuan, as well as along the southeastern coast. All *Cycas* species in China are listed as National First-Class Key Protected Wild Plants under national legislation, reflecting their conservation priority. The southwestern region of China has been identified as a secondary diversification centre for the genus, harbouring substantial morphological and habitat diversity (Xi et al., 2022). However, research on Chinese *Cycas* species has largely focused on taxonomy (Liu et al., 2015; Feng et al., 2016, 2021; Xi et al., 2026), biogeography and phylogenetics (Liu et al., 2021, 2022, 2024), population genetics including large-scale spatial genetic structure (SGS) (Xiao et al., 2004; Feng et al., 2014, 2017; Gong et al., 2015; Yang et al., 2015; Yang et al., 2017; Gong and Gong, 2016; Xiao et al., 2020; Tao et al., 2021; Chang et al., 2022; Tang et al., 2026) and genomic studies (Liu et al., 2022), whereas their fine scale SGS and dispersal patterns are still poorly understood.

Among these, *Cycas szechuanensis* is a critically endangered species endemic to China. It is classified as Critically Endangered (CR) on the Threatened Species List of China's Higher Plants (Qin et al., 2017). This species is currently known to exist in a limited number of wild populations with highly restricted distributions, rendering it particularly vulnerable to habitat fragmentation and anthropogenic disturbances. However, the taxonomic delimitation of this species has a complex history, and the uncertainty surrounding species boundaries has hindered conservation assessments.

Cycas szechuanensis W.C. Cheng & L.K. Fu was first described as a new species in 1975 based on the type specimen (*J.H. Xiong, X.S. Zhang & X.L. Jiang 33221*; holotype: PE; isotypes: HWA, IBK, IBSC, NAS, SZG) collected from a female plant cultivated at Fuhu Temple, Mt. Emei, Sichuan Province, China (Cheng et al., 1975) (Fig. 1a, b). Historical records indicate that this species was originally introduced from Fujian Province (Gong, 2012). *Cycas fairylakea* D. Yue Wang was formally described in 1996 based on a type specimen (*D.Y. Wang & H. Peng 2978*, holotype: SZG; isotype, IBSC, NF) collected from a plant cultivated in the South China Botanical Garden (Wang et al., 1996). Although previously considered as two species, *C. szechuanensis* and *C. fairylakea* are morphologically similar. The primary differences between them are the shape and lobation of the megasporophylls. The apical lobe of megasporophylls is almost indistinguishable from the lateral lobes in *C. szechuanensis*, but it is much more prominent in *C. fairylakea*. Given the overall morphological similarity and relatively minor differences in megasporophylls of several *Cycas* species, the *Cycas taiwaniana* complex has been proposed to include *C. szechuanensis*, *C. fairylakea*, *C. changjiangensis*, *C. hainanensis*, and *C. lingshuigensis*. Subsequent research employing molecular markers to investigate the phylogeography and phylogeny of the *C. taiwaniana* complex synonymised *C. fairylakea* with *C. szechuanensis* (Zhang, 2013; Li, 2016; Wang, 2019; Feng et al., 2021).

Although *C. szechuanensis* type materials were originally collected from Sichuan Province, its largest wild population is primarily distributed in Guangdong Province (Fig. 1a). For example, thousands of individuals have been documented in Shenzhen. Two newly identified populations have been found in Taishan (TS) (>1000 clumps, ca. 100 km^2) and Yangjiang (DB) (>2000 clumps, ca. 50 km^2), both of which are in Guangdong Province. A small wild population comprising approximately 30 clumps has also been discovered in the region of Xinzhou and Beidou Town (XZ) between the TS and DB populations.

Our previous study showed that the two newly identified *C. szechuanensis* populations (TS and DB) are characterised by age structures dominated by juveniles and a balanced sex ratio, whereas adult individuals are scarce and have a slightly female-biased sex ratio (Liang et al., 2024). Field investigations have shown that *C. szechuanensis* seeds are large and heavy and typically fall and germinate in close proximity to the maternal plant due to gravity, resulting in small-scale individual clustering. This aggregated distribution pattern and its potentially limited pollen dispersal (Yang et al., 2024) may restrict gene flow and contribute to the formation of a strong fine-scale genetic structure (FSGS) in *C. szechuanensis*; however, this has not yet been determined. Investigating FSGS in cycad populations under natural field conditions provides valuable insights for management interventions for reducing the aggregation of genetically similar individuals.

Previous studies have reported that *C. szechuanensis* populations exhibit low to moderate genetic diversity and significant inter-population differentiation (Jian et al., 2006; Wang et al., 2006, 2014). However, due to advances in experimental methods and the inclusion of newly discovered populations, these conclusions need to be re-evaluated. The previously used dominant markers (e.g., amplified fragment length polymorphisms and inter-simple sequence repeats) do not assess heterozygosity (Jian et al., 2006; Wang et al., 2006), and use of only three microsatellite markers (also known as simple sequence repeats, SSRs) provides limited genetic resolution (Wang et al., 2014). Moreover, restricted sampling, which was limited to a few small populations in previous research does not allow for a comprehensive understanding of this species (Jian et al., 2006; Wang et al., 2006, 2014; Feng et al., 2021).

In this study, considering improved methodologies and the inclusion of two additional large populations, the genetic diversity of *C. szechuanensis* was reassessed using thorough sampling and mapping in wild and cultivated populations. FSGS was also documented in *C. szechuanensis* for the TS and DB populations. Coupled with parentage analysis, we characterised pollen- and seed-mediated gene flow, which has not been previously reported in *C. szechuanensis* and only seldom reported in other cycads. Based on this comprehensive and systematic analysis, evidence-based recommendations were developed for *C. szechuanensis* conservation.

2. Materials and methods

2.1. Sample collection, DNA extraction, and SSR amplification

Comprehensive field surveys were conducted between 2022 and 2024 to investigate the distribution of wild *C. szechuanensis* populations in Guangdong, Guangxi, and Fujian Provinces, China (Fig. 1a and Table 1). Across all potential distribution regions,

622 *C. szechuanensis* individuals were sampled for population genetic analyses. In addition, 48 *C. taiwaniana* individuals were collected from Fujian and Hainan as outgroups. For the two newly identified populations (TS and DB), more individuals were sampled (TS: n = 421; DB: n = 1257) for FSGS and parentage analysis, and the sex of each individual was determined following the method described by Liang et al. (2024).

Total genomic DNA was extracted from silica gel-dried leaves using a commercial plant DNA extraction kit (Tsingke Biotechnology, Beijing, China). Fifteen polymorphic and reproducible SSR loci were selected for PCR amplification (Table 2). PCR amplification was performed in a 20-μL reaction volume using commercial master mix (Tsingke Biotechnology), with the following cycling conditions: initial denaturation at 98°C for 2 min; 30 cycles of 98°C for 10 s, locus-specific annealing for 10 s and 72°C for 30 s; and final extension at 72°C for 5 min. PCR products were initially verified on 1.5% agarose gel and diluted appropriately for fragment analysis. Genotyping was conducted using capillary electrophoresis on an ABI 3730xl DNA Analyzer (Applied Biosystems, Foster City, CA, USA), with GeneScan™ 500 LIZ® as the internal size standard.

2.2. Data analysis

Null allele frequencies were first estimated using CERVUS 3.0 (Kalinowski et al., 2007) to ensure the reliability of microsatellite genotypes (Table 2). As estimates of null alleles are sensitive to the population composition, we implemented a standardised filtering strategy to maintain consistency across all downstream analyses. Loci with null allele frequencies exceeding 0.20 (Wen et al., 2013) were excluded, yielding 9 loci, which were used for all main analyses, including population-level genetic structure, FSGS, and parentage analyses. The TS and DB populations were analysed separately for FSGS and parentage analyses.

2.2.1. Population genetics of all populations

Standard genetic diversity indices, including the number of private alleles (P_A), observed number of alleles (*Na*), effective number of alleles (*Ne*), observed heterozygosity (*Ho*), expected heterozygosity (*He*), Shannon's information index (*I*), percentage of polymorphic loci (%*P*), population differentiation coefficient (F_{ST}), and inbreeding coefficient (F_{IS}), were calculated using GenAlEx 6.5 (Peakall and Smouse, 2012). Hierarchical analysis of molecular variance (AMOVA) was conducted in Arlequin 3.5 (Excoffier and Lischer, 2010) to partition the genetic variation among and within populations. Principal coordinate analysis (PCoA) was performed using GenAlEx 6.5 to visualise genetic relationships among individuals and populations based on a standardised genetic distance matrix (Peakall and Smouse, 2012). Bayesian clustering was implemented in STRUCTURE 2.3.4 (Evanno et al., 2005) under the admixture model to predict genetic groups in *C. szechuanensis*. *Cycas taiwaniana* samples were excluded to mitigate potential outgroup bias. The number of groups (*K*) was evaluated and set as 1 – 20, with 10,000 burn-in iterations followed by 100,000 Markov chain Monte Carlo (MCMC) iterations. The optimal *K* was determined based on the Δ*K* method using StructureSelector (Kopelman et al., 2015; Li and Liu, 2018).

Bottleneck effects at the population level were assessed using BOTTLENECK 1.2.02 (Piry et al., 1999). Analyses were conducted under the stepwise mutation and two-phase mutation model, with heterozygosity excess and allele frequency distribution tests performed for each model. For each mutation model, the sign and Wilcoxon signed-rank tests were applied. The mode-shift indicator was used to examine the allele frequency distributions; a shifted mode (i.e., an S-shaped distribution), rather than a typical L-shaped distribution, suggests that a population has experienced a recent bottleneck followed by migration or mutation events that contribute to the recovery of genetic diversity. In contrast, an L-shaped distribution indicates that the population has remained in mutation–drift equilibrium and not undergone a recent bottleneck. Cultivated populations were excluded from these analyses.

First-generation migrants were identified using GeneClass2 (Piry et al., 2004) based on the Bayesian criterion of Rannala and Mountain (1997), and recent gene flow was estimated using BayesAss 3.0 (Wilson and Rannala, 2003). MCMC analysis in BayesAss was run for 10,000,000 iterations, with a burn-in of 2000,000 iterations, and samples were recorded every 1000 iterations. A neighbour-joining (NJ) dendrogram was constructed based on Nei's unbiased genetic distances among populations using POPULATIONS 1.2 (Langella, 2000) and visualised using the Interactive Tree Of Life (iTOL; https://itol.embl.de/).

2.2.2. Fine-scale spatial genetic structure and parentage analyses in two newly identified populations (TS and DB)

Fine-scale spatial genetic structure was investigated using two approaches: spatial autocorrelation analysis implemented in GenAlEx 6.5 (Peakall and Smouse, 2012) and estimation of the *Sp* statistic (Vekemans and Hardy, 2004) using SPAGeDi 1.5 (Hardy and Vekemans, 2002). Individuals from the two newly identified populations were classified into two groups: juveniles and adults (Table 3) based on the stem length, representing different age classes (Liang et al., 2024). Spatial autocorrelation analysis was conducted across 26 distance classes in 10-m intervals. The statistical significance of autocorrelation coefficients was assessed in GenAlEx 6.5 using 999 permutations, and confidence intervals were estimated in SPAGeDi 1.5 using 1000 bootstrap replicates. The significance of the *Sp* statistic was evaluated using 1000 permutations. To determine whether the spatial genetic structure differed among sexes and life stages, we performed sex-specific FSGS analysis separately for adult females, adult males, and juvenile females and males in each population.

Parentage analysis was performed for the two newly identified *C. szechuanensis* populations using CERVUS 3.0.7 (Kalinowski et al., 2007) based on a maximum-likelihood approach. In the TS population, individuals identified as artificially transplanted were excluded to ensure that estimates of seed and pollen dispersal distances reflected natural processes. The logarithm of the odds (LOD) threshold for parent–offspring assignments were determined from simulations with 10,000 offspring, assuming a candidate parent sampling rate of 100%, genotyping error rate of 0.01, and minimum typing proportion of 0.95. Parentage assignments were at the 95% strict confidence level. For pollen dispersal analysis, the dispersal distance was calculated as the straight-line distance between the assigned

Table 1
Cycas Species in Guangdong, Fujian and Hainan used in this study.

No.	Group	Population	Location	sample	Origin	Gender
1	FAISE	TS	Taishan, Jiangmen, Guangdong Province	54(421)	Wild	♀♂
2	FAISE	TLS	Tanglang Mountain Country Park, Shenzhen, Guangdong Province	18	Transplanted	♀♂
3	FAISE	HS*	Heshan, Jiangmen, Guangdong Province	2	Wild Remnant	♀
4	FAISE	HJ*	Huaiji, Zhaoqing, Guangdong Province	2	Transplanted	♀♂
5	FAISE	YF*	Yuncheng, Yunfu, Guangdong Province	2	Wild Remnant	♀
6	FAISE	CW*	Longxu District, Wuzhou, Guangxi Zhuang Autonomous Region	20	Wild-origin, clonal	♀
7	FAISE	XZ	Beidou Town, Taishan, Jiangmen, Guangdong Province	29	Wild	♀♂
8	FAISW	DB	Yangdong District, Yangjiang, Guangdong Province	200(1257)	Wild	♀♂
9	FAISW	EHZ	E'huangzhang, Yangchun, Guangdong Province	34	Transplanted	♀♂
10	FAIN	DHS	Dinghu Mountain, Zhaoqing, Guangdong Province	11	Wild Remnant and introduced	♀♂
11	FAIN	LK*	Luokeng, Qujiang, Shaoguan, Guangdong Province	25	Wild-origin, clonal	♀
12	FAIN	DXS*	Danxia Mountain, Renhua, Shaoguan, Guangdong Province	63	Wild-origin, clonal	♀
13	FAIN	LC*	Yangdong Mountain, Lechang, Guangdong Province	40	Wild-origin, clonal	♀
14	FAIN	DP*	Dabu County, Meizhou, Guangdong Province	5	Cultivated	♀
15	FAIN	MK*	Makeng village, Zhao'an County, Zhangzhou, Fujian Province	17	Wild-origin, clonal	♀
16	FAIN	GZ*	Guangzhou, Guangdong Province	3	Cultivated	♀
17	SZE	PY*	Pingyuan County, Meizhou, Guangdong Province	13	Wild-origin, clonal	♀
18	SZE	CSGZ*	Guangzhou, Guangdong Province	13	Cultivated	♀
19-CS	SZE	CSNX*	Nanxiong, Shaoguan, Guangdong Province	1	Cultivated	♀
19-CS	SZE	CSMX*	Wengong Ancestral Hall, Meixian District, Meizhou, Guangdong Province	2	Cultivated	♀
19-CS	SZE	CSLY*	Xueyun Temple, Longyan, Fujian Province	1	Cultivated	♀
19-CS	SZE	CSYA*	Binglü Mountain, Yong'an, Fujian Province	2	Cultivated	♀
19-CS	SZE	CSSM*	Ruiyun Mountain, Chenda Town, Sanming, Fujian Province	14	Cultivated	♀
19-CS	SZE	CSSX*	Taojin Mountain, Shaxian District, Sanming, Fujian Province	35	Cultivated	♀
19-CS	SZE	CSSW*	Yanfang, Nakou Town, Shaowu, Fujian Province	2	Cultivated	♀
19-CS	SZE	CSYP*	Xiyuan Temple, Yanping District, Nanping, Fujian Province	11	Cultivated	♀
19-CS	SZE	CSYT*	Geling Town, Yongtai County, Fuzhou, Fujian Province	1	Cultivated	♀
19-CS	SZE	CSGG*	Guangong Village, Xinxian Town, Putian, Fujian Province	2	Cultivated	♀
20-CT	TAI	CTPTC*	Pingtang Village, Daxi Town, Pinghe County, Zhangzhou, Fujian Province	1	Cultivated	♀
20-CT	TAI	CTXC*	Chen Yuanguang Temple, Xiangcheng District, Zhangzhou, Fujian Province	2	Cultivated	♀
20-CT	TAI	CTHX*	Hexi Village, Yunxiao County, Zhangzhou, Fujian Province	3	Cultivated	♀
20-CT	TAI	CTCT*	Houfang Village, Changtai District, Zhangzhou, Fujian Province	4	Cultivated	♀
21-CH	TAI	CHGZ*	Guangzhou, Guangdong Province	3	Cultivated	♀
21-CH	TAI	CHSY	Sanya, Hainan Province	34	Wild	♀♂

Note: * means the cultivated populations. For conservation purposes, the latitude and longitude of each population are not published. ♀♂ means these populations have female and male individuals, ♀ means these populations only have female individuals, which imply the asexual reproduction. Populations labeled as CS, CT, and CH represent pooled operational populations. Due to negligible genetic differentiation, individuals were grouped into a single population unit to ensure sufficient statistical power for downstream genetic analyses.

Table 2
15 microsatellite primers and null allele frequency in this study.

Locus	GeneBank No.	Primer	Motif	Tm (℃)	Null allele frequency of all pop	Reference
Cha-estssr04	DR063107	F: GATGTTCCCAAATAATGTTACA R: CAAGCTGCACATGCAATGA	(AT)3GT(AT)9AG(AC)4	54	0.4538	(Wang et al., 2008)
CY226	-	F: ACAGGGCATCGGAACACTAC R: CTACTCTTCGGCTTCCAACG	(TA)9	55–53	0.2594	(Cibrián-Jaramillo et al., 2008)
Cpz24	EX809270.1	F: ATGATTCCCATCCAAGC R: TTCCACCCACCACCTATT	(TG)21(GA)18	52	0.3538	(Zhang et al., 2010)
Cy-TaiEST-SSR01	EX921265	F: CGAGAAGTAATTTGCAAATGC R: TGTGAAGCTAAATAGTTGGG	(TC)24(TG)5	55	−0.0744	(Ju et al., 2011)
HL08	EU791556	F: AAAACATTCCTTGCCCTGT R: GGAGCCTGTTGAAGAGCTA	(TTC)12	56	−0.0466	(Li et al., 2009)
CY266	-	F: AAATGCTTTGATGTTCCCAAA R: AAATGCTTTGATGTTCCCAAA	(AT)9	64–54	0.7161	(Cibrián-Jaramillo et al., 2008)
Cha06	EU795702	F: GGTGCATGTGGTGAATAGCC R: GGGAAGGAAAGAGGACAATTA	(CT)21	60	−0.0045	(Zhang et al., 2009)
HL03	EU791551	F: GAGATAGGCTTGGAAGCTTAT R: GTCATAGCCCTCTCTAACAT	(AG)23	60	0.0089	(Li et al., 2009)
Cpz22	EX919867.1	F: CCTCCCAATCCTTCACAAA R: CACCCACAGACAGACCAAT	(GAG)4(GAA)4	52	0.0682	(Zhang et al., 2010)
Cha08	EU827617	F: CAGGGACCATTGTTTCTAAGG R: ACTTATACATAGGGCTCTAAT	(AG)10	54	−0.0047	(Zhang et al., 2009)
CY270	-	F: CGGATTTGGAGGTTCAAAGA R: CAGTTTGTATAGCTGAACAAGAATAGA	(TG)11	55–53	0.3452	(Cibrián-Jaramillo et al., 2008)
Cha03	EU795698	F: TCAAATAATAGTGCCTAGAACC R: AAGAGGCCATCTTGTCTC	(AC)15	60	0.1777	(Zhang et al., 2009)
Cpz2	EX929469.1	F: CTTACGCATACGCCACTTC R: GGACACAACCCCAATCAAA	(TC)10(CA)4(AT)4	52	0.2944	(Zhang et al., 2010)
CY232	-	F: TCTTGCTTACCCGTTTGCTT R: CTCCTCGACGTTCAATCACA	(GT)9(GCGT)3	56–54	0.1616	(Cibrián-Jaramillo et al., 2008)
Cy-TaiEST-SSR11	DR062467	F: GATATTAAAGGCACGGGAG R: TGAAGCTGCTGCATTTGCAT	(CAG)34	56	0.0549	(Ju et al., 2011)

pollen donor and the known mother tree. The seed dispersal distance was calculated as the distance between the known mother tree and the offspring location.

3. Results

3.1. Genetic diversity and population differentiation

3.1.1. Genetic diversity

Based on 9 microsatellite loci (Table 2), genetic variation was assessed across 19 *C. szechuanensis* populations. The newly discovered wild DB population exhibited the highest genetic diversity, with an average *N*a of 7.533, *H*e of 0.555, 100% polymorphic loci, and a relatively low inbreeding coefficient of 0.136. In contrast, cultivated population CSGZ exhibited the lowest diversity ($Na = 1.467$, $He = 0.233$, $\%P = 46.67\%$, and $F_{IS} = -1.000$). All cultivated populations had a negative F_{IS}, indicating excess heterozygosity (Table 3).

Of the two newly identified populations, the DB population had a numerically higher genetic diversity than the TS population in terms of P_A and *N*a, but the differences were not statistically significant. In both populations, juveniles exhibited a numerically higher genetic diversity than adults; however, significant differences were only detected for private alleles, which were significantly more numerous in juveniles than in adults (TS: $P = 0.011$; DB: $P = 0.008$) (Fig. 2a and Table 3). Juveniles in the TS population had a numerically higher *N*a and F_{IS} than adults, but these differences were not statistically significant. This age-related pattern was not consistently replicated when adults and juveniles were separately compared within each sex. In the two populations, juvenile females and males had significantly more private alleles than their adult counterparts ($P < 0.05$). Among TS juveniles, females carried significantly more private alleles than males ($P < 0.05$). For all other genetic diversity metrics, no significant differences were detected between the two populations when comparing the same sex–age classes (Table S1).

3.1.2. Population differentiation and bottleneck effects

Pairwise F_{ST} values and genetic distances among populations are shown in Fig. 2b, c. Populations with sample sizes of fewer than 10 individuals were excluded from this analysis to ensure statistical robustness. Very low pairwise F_{ST} values (approaching zero) were observed in the FAIN (including LK, DXS, LC, and MK) and SZE groups (including PY, CSGZ, and CS) (Fig. 2b), which were identified using structure analysis, as described below (Fig. 1e). The highest pairwise F_{ST} value was observed between FAIN and SZE groups ($F_{ST} = 0.259$). Nei's unbiased genetic distances were highly consistent with the observed F_{ST} patterns (Fig. 2c).

Table 3
Summary of genetic diversity and fine-scale genetic structure of *Cycas szechuanensis* based on 9 SSR loci. Statistics are also reported for adults and juveniles.

Populations	N	P_A	*Na*	*Ne*	*I*	*Ho*	*He*	F_{IS}	%*P*	*F*1	*b*-log	*Sp*
TS	54	0.111	6.267	3.696	1.272	0.500	0.607	0.147	100.00%			
TS-all	421	-	7.389	3.741	1.266	0.593	0.612	0.048	100.00%	0.108***	−0.010	0.011
TS-adults	54	0.222	6.444	3.766	1.251	0.58	0.602	0.055	100.00%	0.123***	−0.012	0.014
TS-juveniles	337	2.111	8.333	3.716	1.28	0.605	0.622	0.041	100.00%	0.110***	−0.012	0.013
TLS	18	0.111	3.067	2.042	0.714	0.389	0.406	0.031	80.00%			
CW*	20	0.444	3.267	1.773	0.630	0.478	0.354	−0.264	93.33%			
XZ	29	0.111	5.267	2.828	1.152	0.561	0.581	0.026	100.00%			
FAISE	**127**	**1.444**	**8.000**	**3.798**	**1.344**	**0.590**	**0.637**	**0.080**	**100.00%**			
DB	200	0.778	7.533	2.904	1.114	0.480	0.555	0.136	100.00%			
DB-all	1257	-	6.778	2.420	0.964	0.539	0.512	−0.033	100.00%	0.120***	−0.024	0.027
DB-adults	200	0.333	5.778	2.397	0.96	0.57	0.513	−0.086	100.00%	0.121***	−0.026	0.029
DB-juveniles	1057	2.333	7.778	2.444	0.968	0.508	0.511	0.021	100.00%	0.121***	−0.024	0.027
EHZ	34	0.444	4.267	2.231	0.921	0.338	0.484	0.339	100.00%			
FAISW	**234**	**1.333**	**6.889**	**2.676**	**1.051**	**0.547**	**0.538**	**−0.001**	**100.00%**			
DHS	11	0.000	3.467	2.256	0.910	0.382	0.513	0.180	100.00%			
LK*	25	0.000	1.933	1.631	0.454	0.571	0.307	−0.790	66.67%			
DXS*	63	0.000	2.933	1.752	0.527	0.573	0.330	−0.660	73.33%			
LC*	40	0.000	2.267	1.567	0.429	0.526	0.287	−0.554	73.33%			
MK*	17	0.000	1.800	1.553	0.410	0.545	0.285	−0.791	66.67%			
FAIN	**164**	**0.000**	**3.444**	**1.734**	**0.560**	**0.649**	**0.362**	**−0.492**	**100.00%**			
PY*	13	0.000	1.533	1.464	0.332	0.451	0.236	−0.815	53.33%			
CSGZ*	13	0.000	1.467	1.467	0.323	0.467	0.233	−1.000	46.67%			
CS*	71	0.000	1.667	1.470	0.332	0.465	0.236	−0.799	53.33%			
SZE	**97**	**0.000**	**1.778**	**1.668**	**0.465**	**0.661**	**0.334**	**−0.980**	**66.67%**			
CT*	10	0.333	1.533	1.502	0.359	0.496	0.256	−0.911	53.33%			
CH	38	3.111	7.000	3.340	1.225	0.454	0.554	0.160	100.00%			
TAI	**48**	**3.667**	**7.889**	**3.643**	**1.218**	**0.526**	**0.517**	**0.018**	**100.00%**			
Total	670	0.259	3.013	1.962	0.619	0.472	0.359	−0.318	73.02%±3.70%			

Note: * means the cultivated populations. N = sample size, P_A = Private alleles, *Na* = mean of observed number of alleles, *Ne* = effective number of alleles, *I* = Shannon index, *Ho* = observed frequency of heterozygotes, *He* = expected frequency of heterozygotes, F_{IS} = coefficient of inbreeding, %*P* = percentage of polymorphic loci. *F*1, means kinship of the first distance class, b-log, regression slope of kinship (*p < 0.05, **p < 0.01, ***p < 0.001). Extent (m) is the extent of FSGS in meters estimated as the distance where *F*ij was significant with the even sample sizes method.

Analysis of molecular variance across the 21 populations revealed that 31.39% of the total genetic variation was partitioned among populations, while 90.57% resided within individuals. Variation among individuals within populations was estimated as −21.96% (Table S2). Most populations showed no significant evidence of a recent bottleneck, with the exception of TLS (Table S3).

3.1.3. Population structure

Bayesian clustering analysis implemented in STRUCTURE identified an optimal number of clusters at $K = 4$. However, K values of 2 – 5 are also presented to better illustrate relationships among populations. At $K = 2$, FAIS (including TS, TLS, HS, HJ, YF, CW, XZ, DB, and EHZ) and SZE clustered together, and FAIN formed a separate group (Figure S1). At K = 3, FAISE and FAISW remained in a single group, and SZE and FAIN formed two distinct groups. At $K = 4$, FAIS was further partitioned into two components: a red component representing FAISE and a purple component representing FAISW (Fig. 1e, f). At $K = 5$, FAISE, FAISW, FAIN, and SZE remained differentiated into four primary groups; however, additional ancestry components were subdivided in FAISW (Figure S1).

Principal coordinate analysis (Fig. 1g) revealed a genetic structure consistent with the STRUCTURE analysis at $K = 4$. Individuals from SZE and FAIN formed distinct groups, and FAISE and FAISW formed two additional groups but with slight overlap. Six wild individuals from Dinghushan (DHS) were assigned to the FAIN group, and five cultivated DHS individuals grouped with FAISE.

In the NJ tree (Figure S2), SZE groups (PY, CS, and CSGZ) formed a well-supported clade positioned between the FAISE and FAIN groups. FAIN constituted a distinct and strongly supported branch, and five cultivated individuals from DHS were nested in FAISE.

3.1.4. Contemporary gene flow and migration

Recent migration rates among the four primary groups (FAISE, FAISW, FAIN, and SZE) were extremely low (Fig. 1h and Table S4). However, asymmetric gene flow was still obvious. Relatively high migration rates were detected from FAISW and FAIN into FAISE with the rates of 0.0274 and 0.0147 respectively, whereas the reverse rates from FAISE into FAIN and FAISW were low (0.0034 and 0.0047).

Interestingly, genetic assignment tests using GeneClass2 indicated that five cultivated individuals from the DHS population were identified as first-generation migrants assigned to FAISE with high confidence ($P < 0.01$), and six wild individuals from DHS were assigned to FAIN (Table S5).

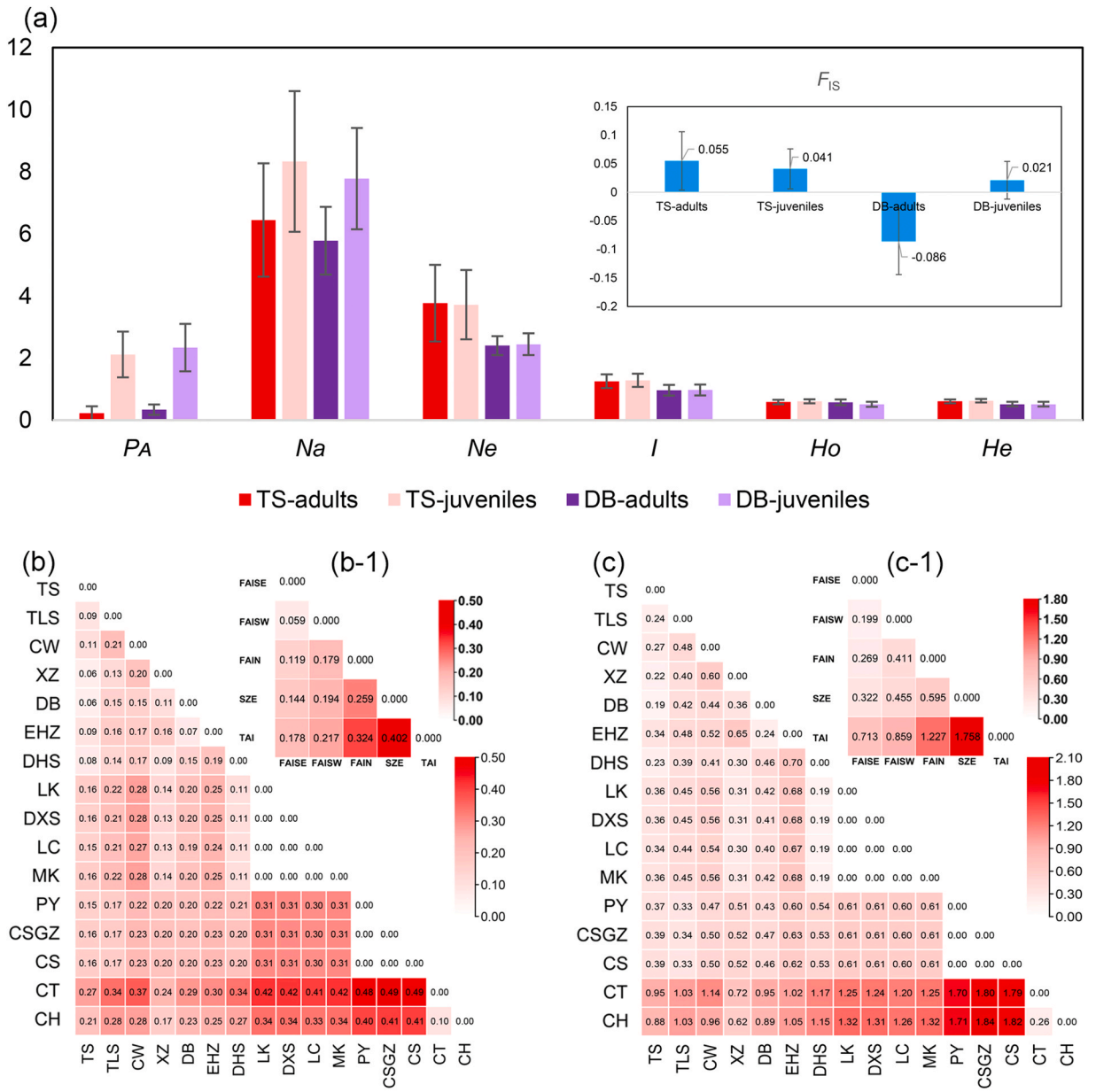


Fig. 2. (a) Genetic diversity parameters of juvenile and adult individuals in the TS and DB populations. Error bars indicate the standard error (SE). (b) Heatmap of pairwise population differentiation (F_{ST}) among the 16 populations (n > 10), including the outgroup, summarised into five genetic clusters (b-1). (c) Heatmap of Nei's genetic distance among 16 populations and (c-1) among the five genetic clusters.

3.2. *Fine-scale spatial genetic structure and parentage analyses in two newly identified populations (TS and DB)*

3.2.1. *Overall FSGS in adults and juveniles*

Significant FSGS was detected in both adults and juveniles in the TS (within 0 – 20 m for adults and 0 – 35 m for juveniles, $P < 0.01$) and DB populations (within 0 – 50 m and 0 – 55 m for juveniles, $P < 0.05$), with a generally similar spatial extent across the two life stages (Fig. 3). Compared to the TS population, the DB population exhibited a higher *Sp* overall, and this pattern was consistent across both adult and juvenile life stages (Table 3).

3.2.2. *Sex- and age-specific FSGS*

In the DB population, a significant spatial genetic structure was detected in adult females, female juveniles, and male juveniles, with a spatial extent of approximately 55 m (Figure S3), indicating that both seed- and pollen-mediated gene flow were locally restricted in the relatively well-preserved DB population. In contrast, pronounced FSGS was detected in the juveniles of both females

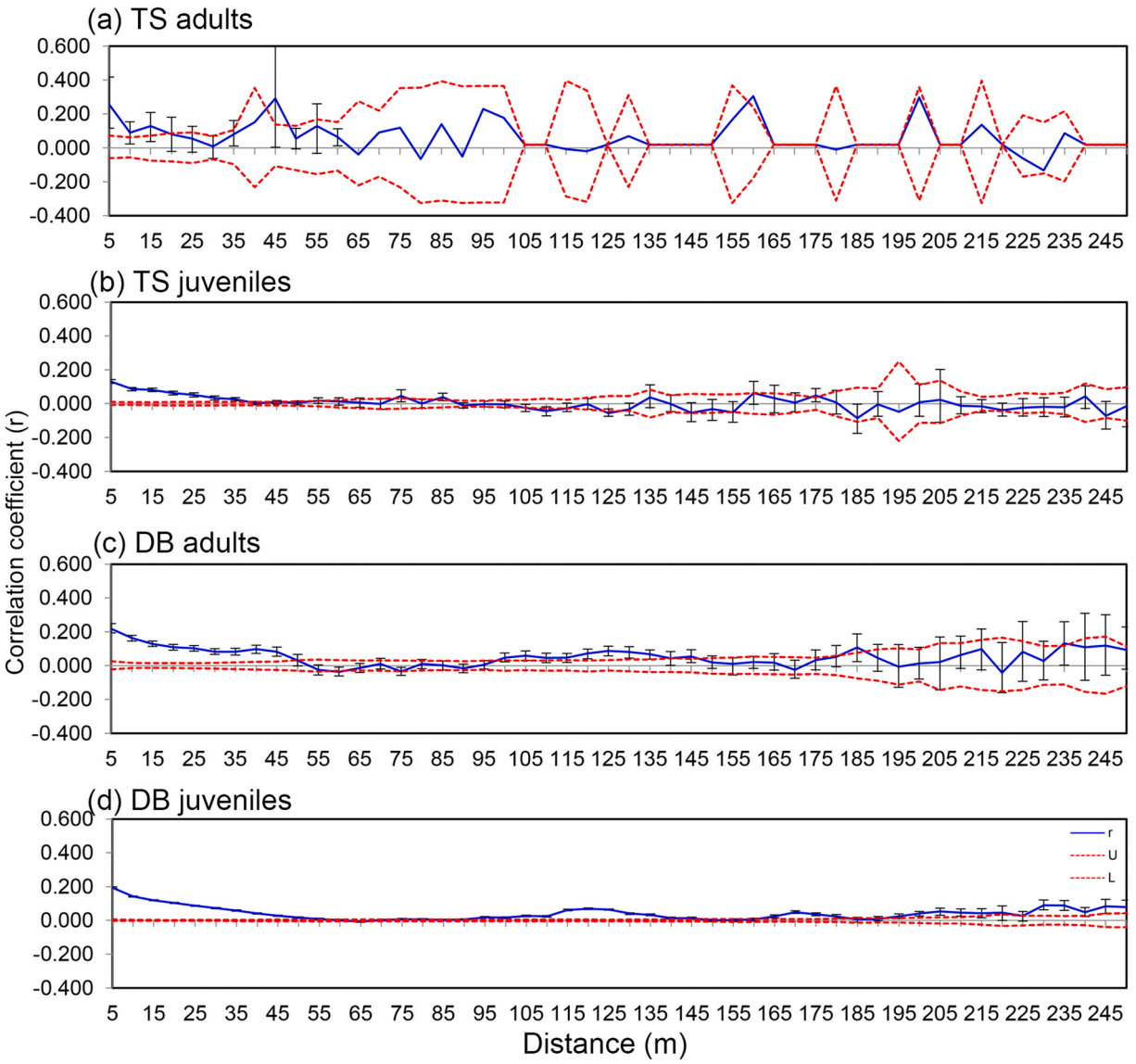


Fig. 3. Fine-scale spatial genetic structure across two newly identified *Cycas szechuanensis* populations in 10-m distance intervals, comprising adults and juveniles. The blue line represents correlation coefficients (*r*) against the geographical distance between individuals with black error bars showing the 95% confidence interval. The red dotted line indicates the upper (*U*) and lower (*L*) limits of the null hypothesis with a 95% confidence interval. A significant spatial genetic structure is inferred when the observed *r* exceeds the upper 95% confidence limit (*U*) of the null distribution. (a) TS adults. (b) TS juveniles. (c) DB adults. (d) DB juveniles.

and males but not in adults of the TS population (0 – 5 m, $P < 0.01$) (Figure S4).

3.2.3. Parentage analysis

At a strict 95% confidence level, parentage assignment rates were low in both TS and DB populations. Among 367 juveniles in the TS population, 69 (18.8%) were assigned to a candidate mother, 25 (6.8%) to a candidate father, and 20 (5.4%) to both parents (i.e., parent pairs). Among 1057 juveniles in the DB population, 68 (6.4%) were assigned to a mother, 48 (4.5%) to a father, and only 6 (0.6%) to a parent pair (Table 4).

Gene flow estimates indicated that the median seed and pollen dispersal distances in the TS population were 9.31 m (mean ± SD: 73.67 ± 225.44 m, range: 0.31 – 1428.9 m) and 91.34 m (mean ± SD: 372.55 ± 466.0 m, range: 3.98 – 1346.9 m), respectively (Fig. 4a and Table 5). In the DB population, the corresponding median dispersal distances were 11.89 m (mean ± SD: 65.28 ± 294.11 m, range: 1.53 – 1746.49 m) for seeds and 21.15 m (mean ± SD: 613.32 ± 930.21 m, range: 6.99 – 1903.67 m) for pollen (Fig. 4b and Table 5).

Table 4
Summaries of the parentage assignment per population.

Populations	TS (9 loci)	DB (9 loci)
Gender	Female/male	Female/male
Parents	27/27	142/58
Saplings	367	1057
Sapling tested	366	1054
Sapling not tested	1	3
Mother only (80%)	190	169
Mother only (95%)	69	68
Father only (80%)	182	168
Father only (95%)	25	48
Parents (80%)	195	32
Parents (95%)	20	6

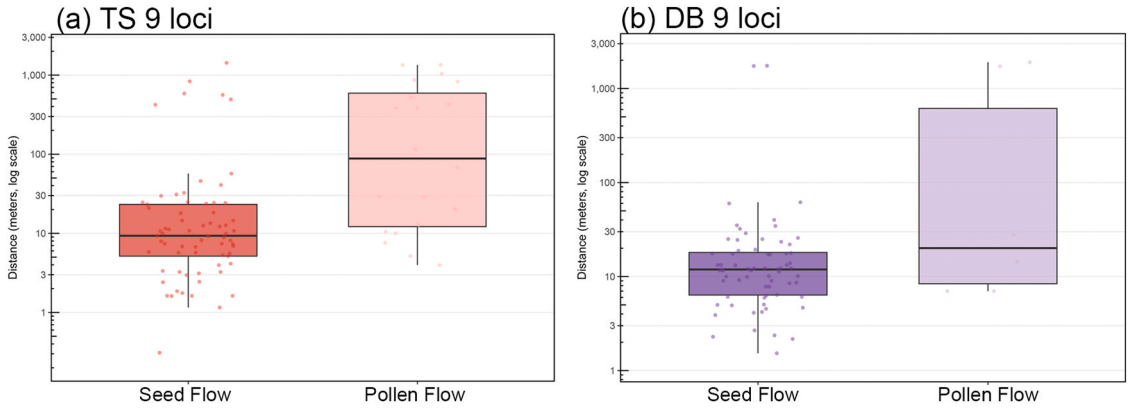


Fig. 4. Distance distribution of seed dispersal and pollen movement detected from maternally assigned (a) TS and (b) DB populations, estimated from parentage assignments at 95% confidence using 9 SSR loci. Boxplots with jittered points show median (centre line), interquartile range (box), and individual dispersal events (points). Distances are $\log_{10}$-transformed for visualization; note the different y-axis ranges between the two populations.

4. Discussion

4.1. Comparison of the genetic diversity between wild and cultivated populations

Building on previous studies (Zhang, 2013; Li, 2016; Wang, 2019; Wang et al., 2019; Feng et al., 2021), our investigation expanded sampling across the species' entire known distribution in Guangdong, Guangxi, and Fujian Provinces, encompassing a more comprehensive set of wild and cultivated populations. Although samples from the type locality of *C. szechuanensis* (Mt. Emei, Sichuan Province, Fig. 1a, b) were not included in this study, previous studies have demonstrated that individuals at this site represent human-mediated transplants and exclusively comprise female plants, with no confirmed male individuals documented since the species description. Furthermore, Feng et al. (2021) showed that the Mt. Emei population is more closely related to cultivated individuals from Fujian and Guangzhou than to other wild populations. Therefore, the absence of samples from the type locality is unlikely to have a significant impact on the conclusions of this study. In the context of broader research on the *Cycas* genus (Wang et al., 2014, 2019; Feng, 2016; Xiao, 2019; Deng et al., 2023) and Zamiaceae family (Meerow et al., 2018; Calonje et al., 2024), our analyses revealed moderate to high levels of genetic diversity in wild populations. These findings contrast with earlier assessments reporting lower diversity ($Na = 1.333 – 2.667$) (Wang et al., 2019) (Table S6; Figure S5). Such discrepancies likely arise from a limited sampling scope as observed in previous studies, which have often focused on isolated population fragments or did not adequately capture the full geographic distribution. By adopting a more comprehensive and representative sampling strategy, our study mitigates

Table 5
Distance of pollen and seed dispersal in two populations.

Populations	Distance of seed dispersal (m)				Distance of pollen dispersal (m)			
	Min	Median	Mean (SD)	Max	Min	Median	Mean (SD)	Max
TS (9 loci)	0.31	9.31	73.67 (225.44)	1428.90	3.98	91.34	372.55 (466.00)	1346.90
DB (9 loci)	1.53	11.89	65.28 (294.11)	1746.49	6.99	21.15	613.32 (930.21)	1903.67

these biases and provides a more accurate assessment of the species' genetic status.

Despite the moderate to high within-population diversity, the four primary genetic groups (FAISE, FAISW, FAIN, and SZE) exhibited pronounced genetic differentiation and extremely low contemporary migration rates (Fig. 1h and Table S4), highlighting a substantial degree of genetic isolation among populations, likely reflecting long-term geographic barriers and habitat fragmentation, which is consistent with findings in other *Cycas* species (Xiao et al., 2004; Feng et al., 2014, 2017).

We also observed that juveniles had significantly more private alleles than adults in the newly identified TS and DB populations (Table 3). The higher number of private alleles in juveniles could reflect ongoing pollen-mediated gene flow from outside the study sites, as external pollen sources can introduce novel alleles into each successive seedling cohort (Mound and Terry, 2001; Braga and Collevatti, 2011; Noreen et al., 2016). However, the reason that the adult individuals carrying these "private" alleles may have already died before we sampled them could not be excluded, since it leads us to conclude that the alleles appearing in their offspring are "private".

Wild populations exhibited evidence of inbreeding. In contrast, cultivated populations showed the lowest genetic diversity, with F_{IS} values approaching −1. This excess heterozygosity indicates asexual reproduction via clonal propagation (Judson and Normark, 1996; Balloux et al., 2003; Feng et al., 2024; Serna-Sánchez et al., 2025). Such a pattern has rarely been reported in other endangered gymnosperms, in which reproductive failure typically results in rapid population decline. However, in these long-lived cycads, populations appear capable of persisting through clonal stasis. Field observations suggest that the loss of reproductive males in fragmented garden populations has constrained sexual reproduction, leading to a reliance on asexual propagation.

We observed a negative variation value among individuals within populations using AMOVA analysis (Table S2). This negative component likely reflects genetic admixture within populations, as detected across the dataset using the structure analysis. This pattern has also been observed in other species, such as *Laurus azorica* reported by Rego et al. (2024) with values from −21.54% to −22.74% due to high levels of genetic admixture.

The origins of the cultivated groups of uncertain provenances warrant separate consideration. The cultivated FAIN group (LK, DXS, LC, DP, MK, and GZ, Table 1) shares a largely similar genetic composition with the wild DHS individuals, supported by the genetic assignment of wild DHS individuals to this FAIN in GeneClass2, suggesting that the former was likely sourced from the latter. Based on the admixture patterns observed at K = 2, SZE (PY, CSGZ, and CS, Table 1) may have been derived from the FAIS group (TS, TLS, HS, HJ, YF, CW, XZ, DB, and EHZ, Table 1). In addition, the cultivated clades (orange SZE group and green FAIN group) likely originated from the clades containing wild populations (red FAISE group and purple FAISW group) (Figure S2). However, in the absence of documented records of human-mediated translocation and given the considerable geographic distance between the two locations, this inference remains tentative.

4.2. Fine-scale genetic structure and gene flow

The spatial distribution pattern of plant populations is primarily shaped by seed dispersal, habitat conditions, and the spatial distribution of the established individuals. In this study, microsatellite markers were employed to characterise FSGS and estimate gene dispersal distances through parentage analysis of adult and juvenile individuals in the newly discovered TS and DB populations. Both populations showed pronounced FSGS, a pattern explained by the seed dispersal biology of *C. szechuanensis*. Like most cycads, this species produces large, heavy seeds that fall and germinate close to the maternal plant. During field surveys we occasionally observed seeds germinating while they remained inside the cone, restricting any chance of seed movement. A similar clumped pattern has been documented in *C. panzhihuaensis*, suggesting that limited seed dispersal is a shared driver of the spatial genetic structure across *Cycas* species (Liang et al., 2024; Yang et al., 2024).

To date, FSGS remains largely undocumented in the *Cycas* genus, with previous studies primarily focusing on large spatial distribution patterns (Yang et al., 2024). Our findings are consistent with those for other gymnosperms. For example, in *Abies alba*, juveniles exhibit significantly stronger FSGS than adults due to restricted seed rain around maternal trees and subsequent density-dependent thinning, with clustered, genetically similar juveniles experiencing higher mortality through intraspecific competition and inbreeding depression (Major et al., 2021). In contrast, *A. nephrolepis* showed an opposite pattern, with larger trees displaying stronger FSGS due to historical demographic processes, such as a past bottleneck event, in which surviving older individuals retained stronger genetic clustering and younger generations gradually eroded FSGS through increased pollen-mediated gene flow, leading to a recommended sampling interval of 30 m for conservation purposes (Kim et al., 2023). Comparable life-stage shifts in FSGS have also been documented in angiosperms but driven by different mechanisms. For example, in the dioecious fig *Ficus cyrtophylla*, strong initial FSGS in juveniles is progressively weakened during later developmental stages due to recruitment dynamics and long-distance, wasp-mediated pollen dispersal (Zhou and Chen, 2010). Conversely, in endangered *Tetracentron sinense*, the FSGS intensity increases from saplings to adults, likely due to high seedling mortality and restricted gene flow imposed by topographic and phenological constraints (Wang et al., 2023).

The attenuation of FSGS from juveniles to adults in both populations is consistent with density-dependent thinning during ontogeny, a common demographic process in which competition-driven mortality reduces the spatial aggregation of related individuals as cohorts mature. However, the decline in the adult population size and the disruption of spatial continuity observed in this study suggest that intensive anthropogenic disturbance, particularly logging for *Eucalyptus* plantations, has likely exacerbated this pattern beyond the level expected from natural thinning alone, threatening the long-term genetic integrity of these populations.

Previous studies have established that cycads in the genus *Cycas* are ambophilous, and wind-mediated pollen dispersal is generally considered to be extremely limited, typically extending no more than 2 – 3 m (Wang et al., 1997; Gan, 2013; Kono and Tobe, 2007). Therefore, insects (particularly beetles and moths) have been regarded as the principal long-distance dispersal agents of *Cycas* (Terry

and Tang, 2009; Cai et al., 2018). Our genetic analysis of *C. szechuanensis*, yields pollen dispersal estimates of up to 1.35 km in TS and 1.9 km in DB, with median distances of 91.34 and 21.15 m. These values are considerably larger than those reported for any other *Cycas* species to date (e.g., *C. revoluta*: effective wind-pollination distance ~2 m, Kono and Tobe, 2007; *C. panzhihuaensis*: ~2.55 m, Wang et al., 1997). One reason is that our genetic estimates capture rare long-distance dispersal events that are difficult to detect in field-based exclusion and pollen-deposition monitoring experiments, which primarily measure short-range patterns (Kono and Tobe, 2007). This limitation is reinforced by the aerodynamic properties of cycad pollen. Although cycad pollen grains are small, they are adhesive and aggregate during descent, with the settling velocity increasing rapidly as the clump size grows. These traits contrast those of typical wind-pollinated plants, whose pollen disperses as dry, non-adhesive single grains and explain why field-based methods have consistently failed to capture long-distance pollen transport in cycads (Hall and Walter, 2011). The low assignment rates indicate that a substantial proportion of pollen and seed dispersal originated from unsampled individuals outside the study plots, consistent with extensive cryptic gene flow.

The observation of long-distance pollen flow carries practical significance for conserving *C. szechuanensis*. If the observed genetic connectivity reflects ongoing pollen-mediated gene exchange among patches, it could buffer the species against genetic drift and inbreeding depression despite limited seed dispersal. Direct observation of pollinators was not conducted in this study, therefore, the actual pollination vectors of *C. szechuanensis* remain to be determined.

5. Conservation recommendations

(1) Immediate protection should be extended to the newly identified wild populations (TS, XZ, and DB). Despite harbouring a relatively high genetic diversity, these populations fall outside existing protected areas, suffer from inbreeding, and remain vulnerable to ongoing habitat loss. The significant FSGS observed in TS and DB populations indicates highly localised seed dispersal and restricted natural recruitment. Therefore, *in situ* conservation areas should encompass sufficient buffer zones to preserve the spatial extent of natural recruitment processes, and *ex situ* collections should be designed using spatially stratified sampling to avoid the over-representation of closely related individuals from within the same genetic cluster.

(2) The low parentage assignment rates and detection of long-distance pollen dispersal (up to ~1.9 km) indicate that gene flow from unsampled individuals and potentially unsurveyed populations substantially contributes to the genetic connectivity of existing populations. Conservation managers should prioritise establishing dynamic buffer zones and local micro-reserves around the inferred pollen dispersal radius to protect undocumented wild remnant individuals from anthropogenic disturbance. Additionally, the habitat corridors that facilitate insect-mediated pollen dispersal should also be preserved and restored to maintain contemporary gene flow among populations.

(3) Carefully managed germplasm exchange between the TS/XZ and DB/EHZ groups is recommended to mitigate moderate lineage divergence and enhance the long-term evolutionary potential. However, such interventions should be implemented cautiously to avoid potential outbreeding depression.

(4) For *ex situ* conservation collections, management practices should transition to a more standardised and genetically informed framework that focuses on distinguishing seed-derived individuals from those propagated clonally via ramets.

(5) In populations restricted to asexual reproduction (e.g., LK, LC, and DXS), we recommend genetic rescue through controlled hand pollination, but need to avoid both the risk of inbreeding depression associated with pollen from the proximate DHS population and outbreeding depression from more distant sources (e.g., TS, XZ, DB, EHZ, and TLS) in the process. All man-made crosses must be rigorously documented, with detailed records of pollen donors, maternal lineages, and progeny identities. Progeny should be monitored for survival, growth, and genetic diversity in a common garden setting prior to reintroduction, ensuring that restoration efforts are grounded in empirical fitness data.

CRediT authorship contribution statement

Dan Liang: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Zheng-Feng Wang:** Writing – review & editing, Supervision, Methodology. **Rui-Jiang Wang:** Writing – review & editing, Supervision, Resources, Funding acquisition.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.gecco.2026.e04441.

Data availability

Data will be made available on request.

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